Calculus II Assignment 5

Part I. Regular Problems

1. Which of the following vectors are unit vectors?

   (a) $\frac{1}{50}(3, 4, 5)$,

   (b) $-\mathbf{U}$ if $\mathbf{U}$ is,

   (c) $(-u_1, u_2, -u_3, u_4, -u_5, u_6)$, if $\mathbf{U} = (u_1, u_2, u_3, u_4, u_5, u_6)$ is.

   (d) $\frac{1}{3}(1, -\sqrt{2}, \sqrt{3}, -\sqrt{3})$.

2. Determine whether the following pairs of vectors are orthogonal to each other.

   (a) $(1, 1, 1, 1, 1)$ and $(-1, -1, -1, -1, -1)$,

   (b) $(1, 1, 1, 1)$ and $(-1, -1, -1, 3)$,

   (c) $(1, 1, 1)$ and $(-1, 2, -1)$.

3. Find equations in the form $ax + by + cz = d$ for the plane through the origin with normal $(1, 0, 0)$ in $\mathbb{R}^3$.

4. Which of these points are on the plane in $\mathbb{R}^3$ given by $20x_1 + 10x_2 - 50x_3 = 0$?

   (a) $(0, 0, 0)$

   (b) $(0, 5, 1)$

   (c) $(-1, 2, 0)$

   (d) $c\mathbf{X}$ if $\mathbf{X}$ is.

   (e)) $\mathbf{X} + \mathbf{Y}$ if $\mathbf{X}$ and $\mathbf{Y}$ are.

5. Which of these points are on the plane in $\mathbb{R}^3$ given by $2x_1 + x_2 - 5x_3 = 100$?

   (a) $(0, 0, 0)$

   (b) $(50, 0, 0)$

   (c) $(0, 100, 0)$

   (d) $(0, 100, 0) + \mathbf{Y}$ if $2y_1 + y_2 - 5y_3 = 0$.

   (e) $\mathbf{X} + \mathbf{Y}$ if $\mathbf{X}$ and $\mathbf{Y}$ are.

Part II. Group Problems

1. Imagine an n-dimensional cube in $\mathbb{R}^n$ with edge length $c > 0$, consisting of all the points $\mathbf{X} = (x_1, x_2, \ldots, x_n)$ with

$$0 \leq x_k \leq c, \quad k = 1, \ldots, n$$

   (a) Find the point $\mathbf{P}$ in the cube that has the largest norm. Call $\mathbf{P}$ the far corner of the cube.

   (b) For what value of c is the far corner $\mathbf{P}$ on the unit sphere of $\mathbb{R}^n$, $\|\mathbf{X}\| = 1$?

   (c) Keeping the far corner point on the unit sphere, what happens to the edge length c as the dimension n tends to infinity?